

David Muñoz del Valle

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EXPERIENCE

Machine Learning and Data Analysis (Python, pandas, PostgreSQL, predictive modeling, CI/CD, MLOps) Mar. 2026 – Present
BHS Corrugated Spain, Production-oriented data analysis and machine learning workflows in an industrial environment Spain

- Extract and analyze industrial data from PostgreSQL with Python and pandas to support operational decision-making and model development.
- Prepare, validate, and engineer features for machine learning use cases, combining data analysis work with production-oriented implementation.
- Build predictive models for industrial scenarios and help move them toward reliable production use through CI/CD-backed workflows.

EDUCATION

B.Eng. in Software Engineering — 7.68 / 10.00 (equivalent GPA: 3.07 / 4.00) 2022 – 2026
University of Malaga Malaga, Spain

- Relevant coursework: Algorithm Analysis and Design, Data Structures, Software Modeling, Design Patterns, Embedded Programming, Backend and Cloud Computing.

PROJECTS

roma Metaheuristic optimization framework (Rust) – zero-dependency design

- Built a 100% Rust optimization framework focused on reusable abstractions and efficient execution.
- Implemented single and multi-objective optimization support in the same library.
- Benchmarked the framework against *pagmo*, the ESA open-source standard, reaching competitive execution speeds.

Brain, AI-assisted chat product backend (Python, FastAPI, Oracle Autonomous Database) – Malackathon II Best Solution Oct. 2025

- Built and deployed a FastAPI backend for a collaborative hackathon project under tight time constraints.
- Integrated the backend securely with Oracle Autonomous Database.
- Worked on the integration layer consumed by a custom frontend for the AI-assisted chat experience.

Dr. Artificial, AI-powered virtual doctor (Python, embeddings, retrieval, web frontend) – 4th Place

Mar. 2025

- Helped build a medical question-answering assistant with a Python backend and LLM-based responses.
- Used embeddings and retrieval workflows to keep multi-step conversations grounded in available clinical information.
- Contributed to an end-to-end product demo combining backend logic and web integration.

PokerHelper, Texas Hold'em desktop decision-support tool (Rust, computer vision, game-state analysis)

In Development

- Building a Rust desktop application to assist decision-making in Texas Hold'em scenarios.
- Combining computer vision with game-state analysis to estimate actions aligned with GTO principles.

WikiMovies, Backend movie database API (Java, Spring Boot, REST API)

- Built a REST API in Java and Spring Boot with a layered architecture separating routing, business logic, and persistence.
- Focused on clean backend structure and maintainable code through explicit separation of concerns.

VMware vSphere ESXi CLI Library, VM lifecycle automation toolkit (Bash, VMware ESXi CLI)

- Built a Bash-based toolkit to provision, clone, and destroy virtual machines directly from the VMware ESXi CLI.
- Focused on infrastructure automation for repetitive VM lifecycle operations in a lightweight command-line workflow.

SKILLS

- **Programming Languages:** Rust, Python, SQL, Java, Bash, C, C++, C#, Haskell
- **AI & Data:** pandas, embeddings, retrieval workflows, feature engineering, predictive modeling, PostgreSQL, data pipelines
- **Backend:** FastAPI, Spring Boot, REST APIs, Oracle Autonomous Database
- **Cloud & DevOps:** Git, GitHub, Linux, Docker, CI/CD, AWS, OCI, Nginx
- **Frontend:** React, TypeScript, Vite, HTML, CSS
- **Speaks:** Spanish (Native), English (Professional), French (Elementary)